

JOYSTICK CONTROL For Upto 5 Appliances For The Disabled

SURESH DWIVEDI

In the world of electronics and embedded systems, the use of a joystick and an Arduino Uno board has many possibilities for developing control systems, especially for disabled individuals with no hands. Creating a joystick with Arduino-based control for five applications can significantly improve the quality of life and independence of the disabled.

The goal of this project is to develop a hands-free control system that allows individuals without hands to independently operate various devices and systems within their homes. By using this joystick with Arduino-based control, users can manipulate the joystick with their elbow, chin, foot, or any other body part to send control signals to the Arduino. The Arduino then sends these signals to switch on or switch off different appliances.

The system allows for the customisation of control parameters, making it adaptable to the user's specific needs and preferences. By providing control over essential home systems, this joystick can enhance the user's comfort and convenience. It can also be expanded to integrate with a wide range of smart home devices, making it a central control hub for multiple applications.

This hands-free smart home control system showcases the potential of joystick and Arduino-based control in improving the daily lives of individuals with physical disabilities, offering them greater independ-

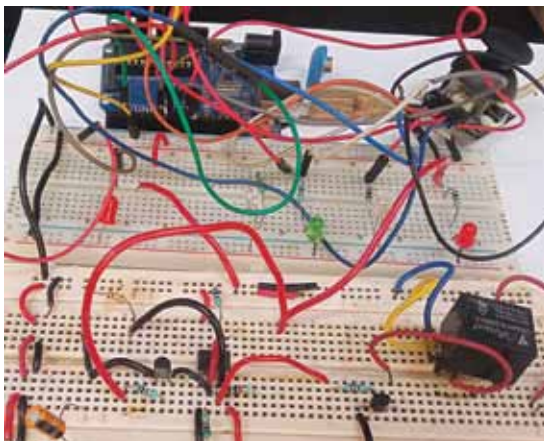


Fig. 1: Author's prototype on bread board

ence and control over their environment. A prototype of the joystick is shown in Fig. 1.

Circuit and working

Fig. 2 shows the circuit diagram of the joystick with Arduino-based control for up to five appliances for the disabled. The circuit is built around a 230V AC primary to 9V, 500mA secondary transformer (X1), a bridge rectifier (BR1), an Arduino Uno board (Board1), a joystick (connected across CON6), 5V voltage regulator LM7805 (IC1), five LM555 timer ICs (IC2 through IC6), five BC547 transistors (T3-T7), five 2N2219 transistors (T1, T2, T8-T10), five 5V SPDT relays (RL1-RL5), the five 230V AC appliances to be controlled (APPL.1 through APPL.5), and a few other components.

The joystick is essentially a combination of two potentiometers for the X and Y planes, respectively. It

reads the voltage through two potentiometers and provides an analogue value to the Arduino, which changes as the joystick shaft moves.

Here we are interfacing the joystick with the Arduino Uno to control five LEDs based on the joystick's movement. This joystick also has a push button that can control one more LED (LED2). Four LEDs (LED3 through LED6)

represent the direction of the joystick shaft movement. A glowing

PARTS LIST

Semiconductors:

IC1	- LM7805, 5V voltage regulator
IC2-IC6	- LM555 timers
T1, T2, T8-T10	- 2N2219 NPN transistors
T3-T7	- BC547 NPN transistors
BR1	- 1A bridge rectifier
D1-D6	- 1N4007 rectifier diodes
LED1-LED6	- 5mm LEDs

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R10, R13-R16, R20-R31	- 1-kilo-ohm
R11, R12, R17-R19	- 100-kilo-ohm

Capacitors:

C1	- 1000 μ F, 35V electrolytic
C2-C6	- 4.7 μ F, 16V electrolytic

Miscellaneous:

CON1-CON5, CON7-CON12	- 2-pin connectors
CON6	- 5-pin connector
BOARD1	- Arduino Uno
RL1-RL5	- 5V SPDT relays
APPL.1-APPL.5	- 230V AC appliances
X1	- 230V AC primary to 9V, 500mA secondary transformer
	- Joystick